

Setting group switch triggered by IEC 61850 GOOSE from:
TR-39 islanding relay $\rightarrow$ Group 1 $\leftrightarrow$ 2 Relay 25 close signal $\rightarrow$ Group 2 $\rightarrow$ 3 $\rightarrow$ 1

Group 1 Grid-Connected

$I_p = 1.50$ pu
TMS= 0.10, SI
Z1/Z2 + 87L active

Islanding detected
(ROCOF/V/f GOOSE)
 $t = 100$ ms

Group 2 Islanded

$I_p = 0.30$ pu
TMS= 0.20, SI
OC primary, 67 dir

Emergency
bypass

Coordination margin at islanded fault level:
Primary (Group 2): 413 ms Backup: 771 ms
Margin: 358 ms > 200 ms requirement

Relay 25 closes
(ΔV , Δf , $\Delta \theta$ OK)

Group 3 Resynchronising

$I_p = 0.80$ pu
TMS= 0.15, SI
25 + V/f monitoring

Grid
restored
(PCC V OK)